

Beef Price Forecasting — Exercise Brief

An exploratory analysis and forecasting exercise on USDA weekly beef price and volume data.

Some exploratory data analysis to get a feel for the series at hand. The accompanying CSV file is used for all data questions.

A. Data Exploration and Cleaning

A.1. Missing values

Examine whether the series has missing values. Which would you discard, and why? Which would you fill, and how? Fill everything at least since 2017 following your chosen method, and discard history before 2017.

A.2. Visualization

Using a simple line plot, chart the three-price series. Are there periods of extreme values? Outside of those periods, is there any visible seasonality? Then focus on the series "Loin, tndrlain, trmd, heavy (189A 4)": does the distribution of values have long tails? How does the price series compare to the volume series of the same cut? Is there any information we can glean from this basic visualization?

A.3. Extreme values

The USDA data has been vetted for outliers, but there are periods where beef prices have been truly extreme. Using basic descriptive statistics, can you hone in on the periods with values that seem extreme or abnormal?

B. Modeling

Construct a basic forecasting model based on linear regression.

B.1. Linear regression with full history

Standardize all series and rename with a **_scaled** suffix. Regress **tndrlain_scaled** against **butt_bnls_scaled**, **butt_CC_scaled**, and an intercept.

- Q1: Is the resulting model better than fitting a constant?
- Q2: Are all regressors significant? Does the answer contradict Q1?

Then drop one of the regressors and fit again.

- Q1: Has the goodness of fit improved, deteriorated, or stayed approximately unchanged?
- Q2: Are the regressors more or less significant than before? (Hint: check how the condition number has changed.)

Create a simple plot of the fit vs. actual.

B.2. Forecasting a single holdout

Use one of the previous models for out-of-sample predictions.

- 1 Split the data into two subsets: before 2021 (train set) and 2021 (test set).
- 2 Fit the OLS model on the train set.
- 3 Predict on the test set.
- 4 Compare to actuals in 2021 to calculate the out-of-sample error per horizon.

C. Cross-validation

How would you document the expected accuracy of the model going forward? Describe a strategy succinctly.